Tang Shaoyan

Technical Practice | Computational Imaging, AI & 3D Interaction

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GitHub: github.com/Zukunft39 | Portfolio: zukunft39.github.io

EDUCATION

National University of Singapore (NUS) Aug. 2026 - Feb. 2028 (Expected)
Master's Student, Computer Engineering

University of Electronic Science and Technology of China Sep. 2022 - Jun. 2026
Bachelor of Engineering in Software Engineering

TECHNICAL PROJECTS & COURSEWORK

VRMicroscope | Computational Imaging & AI Jul. 2024 - Oct. 2026
Unity / C# / Python | Educational VR simulation

- Implemented a separable radix-2 2D FFT, Hann windowing and log-magnitude normalization to visualize spatial-frequency spectra from specimen textures.
- Modeled SIM frequency mixing with shifted complex spectra, bilinear sampling and pupil support; visualized coverage across three orientations for teaching.
- Built a Python HTTP backend with DeepSeek and Groq/Whisper APIs; validated structured responses against scene state for voice guidance, component localization and highlighting.

Image Denoising & Representation Learning Aug. 2026 - Sep. 2026
NUS CEG5304 coursework | NumPy / PyTorch

- Implemented NumPy forward/backpropagation with finite-difference gradient checks, then convolutional autoencoders and denoising autoencoders on Fashion-MNIST.
- Compared clean, masked and Gaussian-noise reconstruction, 16/64-dimensional bottlenecks and frozen-encoder linear probes; analyzed generalization under noise distribution shifts.

Multi-view Reconstruction & Engine Integration Approx. Sep. 2023 - Jan. 2025
Three-person undergraduate team project | Capture, preprocessing and scene/display integration

- Contributed image capture, preprocessing and Unity mesh/point-cloud visualization; investigated PLY import using team outputs from open-source NeRF tools.
- Selected reconstruction targets and prepared RealityCapture/Blender environment assets for team integration into UE/AirSim simulation scenes.

Simulation Control, Memory & Safety Constraints Autumn 2026
NUS CEG5306 coursework | Python / NumPy

- Conducted PPO hover and target-control experiments in gym-pybullet-drones, inspecting trajectories and control errors rather than reward alone.
- Implemented a 2D CBF safety filter and compared memory-based Q-learning, observation-only policies and RNN REINFORCE in partially observable tasks.

TECHNICAL SKILLS

Programming & Tools: Python, C#; JavaScript (web prototypes); JSON, HTTP APIs, openpyxl, Git.

Imaging & AI: FFT/Fourier analysis, SIM principles; NumPy/PyTorch (coursework); LLM/speech APIs.

3D & Engines: Unity; familiar with 3DGS, voxel representations and SfM/MVS through coursework; UE Blueprints for design configuration.

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Planning & Design | Game Systems, Levels & Production Tools

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INDUSTRY EXPERIENCE

Tencent | Arena Breakout Mobile May 2026 - Aug. 2026

Game Product Intern (Product Team 1)

- Maintained event-system configurations across production releases, completing 20+ functional requirements and resolving 90+ bugs through iteration and debugging.
- Rapidly built a browser-based Sokoban demo for an Evita-related event, with level editing, behavior checks and JSON export for design review and client configuration.
- Built a Python/Tkinter/openpyxl workflow for the Girls' Frontline 2 collaboration's out-of-match events, converting structured narrative content into node/dialogue Excel configurations and reducing manual work from tens of hours to minutes.
- Supported regeneration after source updates, protected inputs/existing outputs and checked generated workbooks; documented tools and handover workflows in the project Wiki.

NetEase ThunderFire | FragPunk Mar. 2025 - Jul. 2025

Game System Planning Intern | Combat / Systems

- Configured a character's behavior state machine using the team's Lua toolchain and native UE Blueprints on the design side.
- Configured monetization resources and collaborated on implementation reviews and visual checks to translate design rules into in-game behavior.

GAME DESIGN PROJECTS

Water Adventure | State & Resource Puzzles Oct. 2024 - Nov. 2024

Team game project | Game flow, gameplay and level configuration / scene assembly

- Designed around three water states, form-specific abilities and resource costs, turning transformations into route choices and puzzle conditions.
- Organized levels that progressively introduce interactive objects; prepared level configurations and art requirements, coordinating scene assembly and feedback.

Dreamer | Synchronized Multi-entity Puzzles Nov. 2024

Team Game Jam project | Game flow, levels, scenes and narrative coordination

- Organized light/deep sleep, spirit/physical forms and synchronized entity control into spatial puzzles driven by state changes.
- Configured game flow and levels, assembled scenes and coordinated narrative/resource needs to turn staged rules into playable content.

Mathematical Journey | Interactive Learning Approx. Feb. 2024 - May 2024

Unity 3D team project | Overall design, level / scene assembly and coordination

- Translated traditional mathematics concepts into interactive puzzle objectives; contributed overall flow design and level/scene assembly.
- Coordinated gameplay, art requirements and interaction feedback so that concepts could be understood through player actions.

DESIGN & TOOLS

Design: Systems/gameplay rules, level configuration, scene assembly, state machines, resource configuration and cross-functional reviews.

Prototyping: Unity, native UE Blueprints, Lua toolchain configuration, web/JSON prototypes and Python/Excel automation.